



Trang H. Tran

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EDUCATION

- 2020 – 2024 **Ph.D.**, [Cornell University](#)
School of Operations Research and Information Engineering.
Major: Mathematical Programming.
Advisor: Dr. Katya Scheinberg, Co-advisor: Dr. Lam M. Nguyen.
- 2015 – 2019 **B.S.**, [Hanoi National University of Education](#)
Major: Mathematics and Education.

RESEARCH EXPERIENCE

- 2025 – Present [Postdoctoral Research Associate](#)
Lehigh University, Bethlehem, PA
Supervisor: Dr. Luis Nunes Vicente
- Summer 2024 [Siegel PiTech PhD Impact Fellow](#)
ADAPT Community Network, New York City, NY
Supervisor: Ronak Parikh
- Summer 2023 [Research Scientist Intern \(Artificial Intelligence\)](#)
IBM, Thomas J. Watson Research Center, Yorktown Heights, NY
Supervisors: Dr. Lam M. Nguyen, Dr. Kyongmin Yeo, Dr. Nam H. Nguyen
- Summer 2022 [Research Scientist Intern \(Artificial Intelligence\)](#)
IBM, Thomas J. Watson Research Center, Yorktown Heights, NY
Supervisor: Dr. Lam M. Nguyen

PAPERS

Peer-reviewed Journal Papers

- 2026 [Adjusted Shuffling SARAH: Advancing Complexity Analysis via Dynamic Gradient Weighting](#)
Duc Toan Nguyen, **Trang H. Tran**, Lam M. Nguyen
[Journal of Optimization Theory and Applications \(JOTA\)](#)
- 2025 [Stochastic ISTA/FISTA Adaptive Step Search Algorithms for Convex Composite Optimization](#)
Lam M. Nguyen, Katya Scheinberg, **Trang H. Tran** †
[Journal of Optimization Theory and Applications \(JOTA\)](#)
- 2024 [Shuffling Momentum Gradient Algorithm for Convex Optimization.](#)
Trang H. Tran, Quoc Tran-Dinh, Lam M. Nguyen
[Vietnam Journal of Mathematics \(VJOM\)](#)
Special issue dedicated to Dr. Tamás Terlaky on the occasion of his 70th birthday, 2024

Peer-reviewed Conference Papers

- 2024 [Shuffling Gradient-Based Methods for Nonconvex-Concave Minimax Optimization](#)
Quoc Tran-Dinh, **Trang H. Tran**, Lam M. Nguyen
[Conference on Neural Information Processing Systems \(NeurIPS\)](#) (25.8% acceptance rate)
- 2024 [Improving Time Series Encoding with Noise-Aware Self-Supervised Learning and an Efficient Encoder](#)
Anh Duy Nguyen, **Trang H. Tran**, Hieu H. Pham, Phi Le Nguyen, Lam M. Nguyen
[IEEE International Conference on Data Mining \(ICDM\)](#) (19.5% acceptance rate)
- 2023 [On the Convergence to a Global Solution of Shuffling-Type Gradient Algorithms](#)
Lam M. Nguyen*, **Trang H. Tran***
[Conference on Neural Information Processing Systems \(NeurIPS\)](#) (26.1% acceptance rate)
- 2022 [Nesterov Accelerated Shuffling Gradient Method for Convex Optimization.](#)
Trang H. Tran, Katya Scheinberg, Lam M. Nguyen
[International Conference on Machine Learning \(ICML\)](#) (21.9% acceptance rate)
- 2021 [SMG: A Shuffling Gradient-Based Method with Momentum](#)
Trang H. Tran, Lam M. Nguyen, Quoc Tran-Dinh
[International Conference on Machine Learning \(ICML\)](#) (21.47% acceptance rate)

Peer-reviewed Workshop Papers

- 2023 [Stochastic FISTA Step Search Algorithm for Convex Optimization](#)
Trang H. Tran, Lam M. Nguyen, Katya Scheinberg
Conference on Neural Information Processing Systems (**NeurIPS**),
Workshop on Optimization for Machine Learning (**OPT**)

Preprints Under Review

- 2026 [Stochastic Block Coordinate and Function Alternation for Multi-objective Optimization and Learning](#)
Trang H. Tran, Luis Nunes Vicente †
- 2022 [New Perspective on the Global Convergence of Finite-Sum Optimization](#)
Lam M. Nguyen*, **Trang H. Tran***, Marten van Dijk

Technical Reports

- 2023 [A Univariate Pretrain-Finetune Approach for Time Series Using Supervised Contrastive Learning](#)
Trang H. Tran, Lam M. Nguyen, Kyongmin Yeo, Nam Nguyen, Roman Vaculin
- 2023 [An End-to-End Time Series Model for Simultaneous Imputation and Forecast](#)
Trang H. Tran, Lam M. Nguyen, Kyongmin Yeo, Nam Nguyen, Dzung Phan, Roman Vaculin, Jayant Kalagnanam
- 2022 [Finding Optimal Policy for Queueing Models: New Parameterization](#)
Trang H. Tran, Lam M. Nguyen, Katya Scheinberg

Working Papers

- 2025 [Non-Monotone Stochastic Derivative-Free Optimization Methods](#)
Anjie Ding, **Trang H. Tran**, Luis Nunes Vicente †
- * denotes equal contributions, † denotes alphabetical order

PROFESSIONAL ACTIVITIES

Area Chair

- 2025 [Program Committee – Area Chair \(peer-reviewed conferences\)](#)
Conference on Neural Information Processing Systems (NeurIPS 2026)
International Conference on Learning Representations (ICLR 2026)

Member

- 2022 – Present [Editorial Board, Machine Learning Journal](#)

Organizer

- 2023 [Workshop Organizer](#)
When Machine Learning meets Dynamical Systems: Theory and Applications.
Lam M. Nguyen, **Trang H. Tran**, Wang Zhang, Subhro Das and Tsui-Wei Weng
Workshop at the Conference on Artificial Intelligence (AAAI)
- 2021 – Present [Session Chair](#)
INFORMS Annual Meeting 2025 – "Iterative Optimization Approaches"
INFORMS Annual Meeting 2023 – "First-Order Methods for Machine Learning"
INFORMS Annual Meeting 2022 – "Optimization for Machine Learning"
INFORMS Annual Meeting 2021 – "Recent Advances in Stochastic Gradient Algorithms"

Reviewer

- 2020 – Present [Program Committee – Reviewer \(peer-reviewed conferences\)](#)
International Conference on Machine Learning (ICML 2020 - 2026)
Conference on Neural Information Processing Systems (NeurIPS 2021 - 2025)
International Conference on Learning Representations (ICLR 2021 - 2024)
Conference on Artificial Intelligence (AAAI 2022, 2025)
International Conference on Artificial Intelligence and Statistics (AISTATS 2021 - 2026)
Conference on Uncertainty in Artificial Intelligence (UAI 2022 - 2023)
Conference on Computer Vision and Pattern Recognition (CVPR 2023)

2021 – Present [Reviewer \(peer-reviewed journals\)](#)

Machine Learning (2021 – Present)
Journal of Machine Learning Research (2022 – Present)
Journal of Optimization Theory and Applications (2022 – Present)
Journal of Global Optimization (2025 – Present)
Mathematics of Operations Research (2025 – Present)
Neural Networks (2022 – 2023)
IEEE Transactions on Neural Networks and Learning Systems (2022)
IEEE Transactions on Signal Processing (2021)

2021 – 2026 [Program Committee – Reviewer \(peer-reviewed workshops\)](#)

Principled Design for Trustworthy AI - Interpretability, Robustness, and Safety Across Modalities (Trustworthy AI 2026 – ICLR 2026 Workshop)
Optimization for Machine Learning: Beyond Worst-case Complexity (OPT 2021 – NeurIPS 2021 Workshop)
New Frontiers in Federated Learning: Privacy, Fairness, Robustness, Personalization and Data Ownership (NFFL 2021 – NeurIPS 2021 Workshop)

TEACHING EXPERIENCES

Instructor of Record - Lehigh University

Fall 2025 ISE 364/464 Introduction to Machine Learning (~ 40 students)

Teaching Assistant - Cornell University

Fall 2024 ORIE 6730 Mathematics of Deep Learning (~ 40 students, grading, holding office hours)

Spring 2024 ORIE 3310 Optimization II (~ 90 students, grading, leading discussion)

Fall 2022 MATH 2940 Linear Algebra for Engineers (~ 500 students, grading, leading 3 discussion sections each with 25 students)

Spring 2021 ORIE 3510 Introduction to Engineering Stochastic Processes (~ 100 students, grading)

Mentorship

2024 – Present [Duc Toan Nguyen](#)

Undergraduate student, Mathematics and Computer Science, Texas Christian University. (co-advise with Dr. Lam M. Nguyen)

2022 – 2023 [Anh Duy Nguyen](#)

Ph.D. student, Department of Electrical and Computer Engineering, University of Illinois at Urbana-Champaign. (co-advise with Dr. Lam M. Nguyen)

2022 – 2023 [Angelos Assos](#)

Undergraduate student, Computer Science and Mathematics, Massachusetts Institute of Technology. (co-advise with Dr. Lam M. Nguyen and Prof. Luca Daniel)

GRANTED PATENT

2026 [Training A Neural Network Using an Accelerated Gradient with Shuffling.](#)

Patent US12579434B2
Lam M. Nguyen, **Trang H. Tran.**

PATENT APPLICATIONS

2024 [A Multivariate Time Series Model Which Simultaneously Learns The Spatial And Temporal Relation.](#)

Filed on January 5th, 2024.
Lam M. Nguyen, **Trang H. Tran.**

2023 [Time Series Forecasting Using Multivariate Time Series Data With Missing Values.](#)

Filed on February 1st, 2023.
Lam M. Nguyen, **Trang H. Tran**, Kyongmin Yeo, Nam Nguyen, Dzung Phan, Roman Vaculin, Jayant Kalagnanam.

HONORS & AWARDS

2024 [Cornell Tech PhD Impact Fellowship](#)

Siegel Family Endowment PiTech PhD Impact Fellowship, Summer 2024

2023 [NeurIPS 2023 Scholar Award](#)

Conference on Neural Information Processing Systems (NeurIPS 2023) (Travel Grant)

- 2022 [UAI 2022 Top Reviewer](#)
Conference on Uncertainty in Artificial Intelligence (UAI 2022)
- 2022 [ICML 2022 Participation Grant](#)
International Conference on Machine Learning (ICML 2022)
- 2021 [ICLR 2021 Outstanding Reviewer](#)
International Conference on Learning Representations (ICLR 2021) Reviewer Award (Top 10%)
- 2020 [ORIE Field Fellowship](#)
Eleanor and Howard Morgan PhD'68 Graduate Fellowship, Fall 2020
- 2019 [Young Talent Scholarship Programme 2019](#)
From the Vingroup Innovation Foundation (VINIF) for outstanding students who are pursuing the domestic postgraduate study programmes
- 2016 – 2018 [Students Scholarship in National Program](#)
for the Development of Mathematics until 2020
- 2016 [First Prize on Algebra](#)
In Vietnam Mathematics Competition for University Students (nationwide award)

TALKS

- 04/2026 [Shuffling Stochastic Gradient Methods in Finite-sum Optimization.](#)
Machine Learning Seminar, Department of Mathematics, University of Pittsburgh, PA
- 02/2026 [Shuffling Stochastic Gradient Methods in Finite-sum Optimization.](#)
Department of Combinatorics & Optimization, University of Waterloo, Canada
- 10/2025 [Stochastic Gradient Methods Without Replacement in Finite-sum Optimization.](#)
INFORMS Annual Meeting 2025, Georgia, GA
- 07/2024 [Stochastic ISTA/FISTA Adaptive Step Search Algorithms for Convex Composite Optimization.](#)
International Symposium on Mathematical Programming (ISMP 2024), Montreal, Canada
- 12/2023 [On the Convergence to a Global Solution of Shuffling-Type Gradient Algorithms.](#)
Conference on Neural Information Processing Systems (NeurIPS 2023), New Orleans, LA
- 10/2023 [Stochastic Optimization Methods with Momentum.](#)
INFORMS Annual Meeting 2023, Phoenix, AZ
- 01/2023 [Shuffling Gradient Methods for Finite-sum Optimization.](#)
Institute of Mathematics, Vietnam Academy of Science and Technology, Hanoi, Vietnam
- 10/2022 [Nesterov Accelerated Shuffling Gradient Method.](#)
INFORMS Annual Meeting 2022, Indianapolis, IN
- 07/2022 [Nesterov Accelerated Shuffling Gradient Method for Convex Optimization.](#)
International Conference on Machine Learning (ICML 2022), Baltimore, MD
- 10/2021 [Shuffling Gradient-Based Methods](#)
INFORMS Annual Meeting 2021, Anaheim, CA
- 07/2021 [SMG: A Shuffling Gradient-Based Method with Momentum](#)
International Conference on Machine Learning (ICML 2021), virtual conference

REFERENCES

Katya Scheinberg, Ph.D.

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